

Vincent Schüßler

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Experience

Wētā FX, Visiting Researcher

Jan 2026 – July 2026

- Developing sampling techniques for volumetric path tracing
- Implementation in the Manuka production renderer (C++)

KIT Computer Graphics Group, Researcher

Feb 2019 – Jan 2025

- Conducted and published research projects
- Developed and contributed to research rendering software (Rust, C)
- Advised students for theses and courses

Education

Dr.-Ing. Computer Graphics Group, Karlsruhe Institute of Technology (KIT)

Feb 2019 – July 2026

(PhD) My research focused on the efficient and high-quality rendering of complex light transport phenomena. Other interests include real-time GPU path tracing, texturing, filtering and microfacet models.

- Thesis: *Guided Sampling and Caching for Efficient Physically-Based Rendering*
- Expected completion in July 2026 (defense scheduled)

M.Sc. Computer Science, Karlsruhe Institute of Technology (KIT)

Apr 2016 – Nov 2018

- Thesis: *High-Dimensional Path Guiding* (computer graphics)
- Areas of specialization: Computer graphics and machine learning

B.Sc. Computer science, Karlsruhe Institute of Technology (KIT)

Oct 2012 – Feb 2016

- Thesis: *Geometrically Correct Normal Maps Using Non-Symmetric Microsurfaces* (computer graphics)

Projects

Demoscene activity as member of rtifical

2013 – 2017

- Developed shaders for SDF-based rendering and post processing (GLSL)
- Contributed to custom tools for visual sequencing and real-time music synthesis (C++)

Skills

Programming languages: Rust, Python, C, C++, GLSL, JavaScript

Technologies: Vulkan, Git, Linux, JAX

Languages: English (fluent), German (native)

Publications

Selective Caching in Procedural Texture Graphs for Path Tracing

July 2025

Vincent Schüßler, Johannes Hanika, Basile Sauvage, Jean-Michel Dischler, Carsten Dachsbacher

Eurographics Symposium on Rendering

<https://doi.org/10.2312/sr.20251176>

Sampling of Anisotropic Spatial Gaussians for Path Guiding

May 2025

Sergey Lelyakin, Vincent Schüßler, Carsten Dachsbacher

Eurographics 2025 - Posters

[10.2312/egp.20251017](https://doi.org/10.2312/egp.20251017)

ReSTIR Subsurface Scattering for Real-Time Path Tracing

Aug 2024

Mirco Werner, Vincent Schüßler, Carsten Dachsbacher

Proc. ACM CGIT (Proc. High Performance Graphics)

[10.1145/3675372](#)**Bridge Sampling for Connections via Multiple Scattering Events**

July 2024

Vincent Schüßler, Johannes Hanika, Carsten Dachsbacher

Computer Graphics Forum (Proc. Eurographics Symposium on Rendering)

[10.1111/cgf.15160](#)**Markov Chain Mixture Models for Real-Time Direct Illumination**

July 2023

Addis Dittebrandt, Vincent Schüßler, Johannes Hanika, Sebastian Herholz, Carsten Dachsbacher

Computer Graphics Forum (Proc. Eurographics Symposium on Rendering)

[10.1111/cgf.14881](#)**Path Guiding with Vertex Triplet Distributions**

July 2022

Vincent Schüßler, Johannes Hanika, Alisa Jung, Carsten Dachsbacher

Computer Graphics Forum (Proc. Eurographics Symposium on Rendering)

[10.1111/cgf.14582](#)**Microfacet-based normal mapping for robust Monte Carlo path tracing**

Nov 2017

Vincent Schüßler, Eric Heitz, Johannes Hanika, Carsten Dachsbacher

Transactions on Graphics (Proc. SIGGRAPH Asia)

[10.1145/3130800.3130806](#)